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Batch: April 2026

E-Commerce Customer Churn Prediction and Retention System

Abstract

This report presents a closed-loop Machine Learning (ML) Customer Retention Platform designed to mitigate user churn and maximize customer lifetime value (CLV) for E-commerce, subscription-based, and Direct-to-Consumer (D2C) brands. Unstructured behavioral data pipelines—encompassing 3,046 transaction logs, 3,910 web session inputs, and 603 customer support ticket histories—were synthesized programmatically for a 500-customer test cohort.

A unified production feature matrix was engineered to train an extreme gradient boosting model (XGBoost) for classification alongside a RandomForestRegressor for continuous monetary forecasting. Explainable AI (XAI) capabilities were achieved via Tree-based Shapley Additive Explanations (SHAP), mapping isolated risk drivers for each account.

Departing from traditional passive diagnostic indexes, this platform establishes an execution layer that triggers personalized offers and product recommendations automatically. Users with a predictive risk metric $\geq 70\%$ are immediately queued into automated communication modules with tailored incentives matching their explicit SHAP root causes.

Operational evaluation of the pipeline successfully captured 158 high-risk accounts. Applying an 18% recovery proxy model yielded 28 simulated conversions, successfully preventing significant fiscal leakage by reclaiming projected retained revenue driven directly by ML-based CLV predictions. The results demonstrate that marrying explainable predictive intelligence with context-aware, automated execution bridges the historic gap between diagnostic analytics and tactical business operations.

Introduction

Customer churn represents one of the most critical threats to compounding growth in modern digital retail environments. Acquiring a new user typically demands significantly higher capital allocations than maintaining an active account. For consumer-focused digital platforms, growth optimization depends heavily on tracking operational events across multiple communication networks.

Historically, firms have treated database engineering, predictive statistical modeling, and marketing automation as disjointed departments. This structural isolation leads to significant issues:

- **Data Disconnects:** Raw web traffic and support ticket histories are rarely merged into real-time forecasting pipelines.
- **Passive Tools:** Traditional analytical tools output descriptive statistical charts but fail to trigger operational countermeasures.
- **Delayed Decisions:** Manual database queries by customer support staff cannot keep up with rapid changes in customer behavior.

This project addresses these gaps by establishing a unified pipeline. The system processes raw data inputs, trains dual-tier predictive models, and runs an automated execution framework to intercept churners in real time.

Problem Statement

Modern retention teams are overwhelmed by data but starved for actionable insights. While basic analytics can flag a declining account, existing solutions frequently fail to address three crucial requirements:

[Raw Multi-Stream Logs] —> [Descriptive KPIs Only] —> Missing Automated Action

1. **Lack of Automated Personalization:** Traditional systems do not tailor corporate incentives to a user's unique friction point. Sending a generic coupon to a user experiencing long customer support delays fails to address their core problem.

2. **Failure to Value Financial Impact:** Basic churn scoring systems treat all accounts equally. Without integrated CLV projections, businesses waste marketing capital on low-margin users while losing high-value accounts.
3. **Absence of Straight-Through Processing (STP):** Most analytical platforms function strictly as lookup indexes. They require corporate staff to manually search historical profiles, design campaigns, and handle distribution.

This project aims to fix these shortfalls by engineering an automated system that connects predictive risk modeling directly to targeted email systems and product recommendation matrices.

Results and Discussion

Data Pipeline Consolidation

The programmatic data engine generated multi-stream historical arrays to evaluate the pipeline under realistic conditions:

- **Transaction Records:** 3,046 entries tracking purchase timelines and order margins.
- **Web Telemetry Stream:** 3,910 logs capturing category interest and cart abandonment variables.
- **Customer Support Records:** 603 tickets measuring resolution time and sentiment scores.

These multi-source inputs were aggregated into an analytical feature matrix of shape (500, 12). This stage consolidated transactional features with behavioral indicators, such as the user's primary browsing category.



Machine Learning & XAI Performance

The model utilized an XGBoost Classifier combined with Synthetic Minority Over-sampling Techniques (SMOTE) to correct underlying class imbalances. Tree Explainer loops successfully translated complex model weights into clear business reasons. Rather than generating global metrics, the system isolated individual churn drivers for every customer.

isolated SHAP Risk Driver	Dynamic Promotional Voucher
recency	We miss you! Get a 25% discount on your next checkout.
total_cart_abandoned	Items are waiting! Free Shipping applied to your cart.
avg_sentiment	Priority account manager assigned for 1-on-1 support resolution.

Concurrently, the RandomForestRegressor successfully generated continuous CLV projections. This allowed the system to separate users into value tiers (High Value vs. Low Value) via quantile distribution.

Automated Execution & Revenue Impact

The automated engine successfully identified 158 customers crossing the critical 70% churn risk threshold. Instead of stopping at a manual database lookup, the system ran straight-through processing loops that instantly created 158 personalized retention payloads.

[158 High-Risk Users Identified]



[Apply 18% Retention Conversion Rate]



[28 Saved Accounts] —> [Projected Retained Revenue Reclaimed]

The system mapped each user's primary risk factor to a targeted promotion, bundled it with recommendations matching their favorite product category, and generated a verified communication log. This automation strategy shifts the dashboard from a passive reporting tool into an active driver of business value.

Conclusion

This project demonstrates the deployment of an end-to-end, automated customer retention architecture. By combining behavioral engineering, machine learning classification, and tree-based explainability, the platform successfully automates customer retention operations.

The system transitions dashboards from simple retrospective charts into automated control hubs. By automatically targeting marketing spend at high-value customers based on their specific risk profiles, businesses can lower user acquisition costs and increase lifetime value. This approach provides e-commerce, D2C, and subscription companies with a clear blueprint for data-driven customer retention.

References

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